

UNIVERSITY OF BIRMINGHAM

School of Computer Science

Week 3 Exercise Questions

Artificial Intelligence 2

2024

Artificial Intelligence 2

Exam paper

Question 1 (2021 AI2 Exam question)

As a data scientist in a telecommunication company, your task is to analyse a customer dataset to predict whether a customer will terminate his/her contract. The dataset consists of around 8000 customer records, each consisting of one binary dependent variable Y , indicating whether the customer terminates the contract ($Y = 1$) or not ($Y = 0$), and 19 independent variables, which include the customer's information, e.g., age, subscription plan, extra data plan, etc., and the consumer behaviour such as average numbers of calls and hours per week. Since your boss needs some actionable insights to retain customers, you decided to use interpretable machine learning methods. Design your interpretable machine learning method by answering the following questions:

- (a) You have implemented a feature selection algorithm based on mutual information to select the most informative features from the 19 independent variables. To validate the implementation of your mutual information calculation function, you use a small subset of the data to calculate mutual information manually. You select one independent variable 'subscription plan', denoted as S , which takes two values, $S \in \{1, 2\}$. Please use the following Probability Mass Function table

$p(S, Y)$	$S = 1$	$S = 2$
$Y = 0$	$\frac{2}{12}$	$\frac{5}{12}$
$Y = 1$	$\frac{2}{12}$	$\frac{3}{12}$

to calculate

- Entropies $H(S)$ and $H(Y)$
- Conditional entropies $H(S|Y)$ and $H(Y|S)$
- Joint entropy $H(S, Y)$
- Mutual information $I(S; Y)$

Show all your working. Discuss what mutual information means and whether this feature will be selected or not.

- (b) After applying your algorithm you selected two variables: 1) extra data plan E , which is a binary random variable that indicates whether the customer subscribes to the extra data plan ($E = 1$) or not ($E = 0$); and 2) averaged hours used per week H , which is a continuous random variable. You then built a logistic regression model

to classify customers into 'low risk' or 'high risk' of terminating the contract. The fitted model is

$$\log \left(\frac{p}{1-p} \right) = -0.77 + 0.23H - 1.18E$$

- Given a customer who has the extra data plan ($E = 1$) and spent on average 0.5 hours per week, calculate the odds and the probability the customer will terminate the contract ($Y = 1$). **(4 marks)**
- Using this fitted model, explain to your boss what actions should be taken to retain customers. **(10 marks)**

0 marks for question not valid - all questions must have the same number of marks

Question 2 (2022 AI2 Exam question)

As a machine learning expert for an AI cyber security company, your task is to design an automated network intrusion detection system. You have collected a large number of records of network activities. Each record includes the log information about network activity, such as protocol types, duration, number of failed logins, which are random variables, denoted as $\mathbf{X} = [X_1, X_2, \dots, X_n]^T$. Each record also includes a binary random variable Y called label that was labelled by cyber security experts as intrusions ($Y = 1$) or normal connections ($Y = 0$).

- (a) Consider feature selection based on mutual information to reduce the number of independent variables.
- (i) Explain to your colleague, who knows nothing about information theory, the concept of mutual information. **[2 marks]**
- (ii) Explain the loop, i.e., lines 4-7 of the pseudocode in Table 1. Note $I(Y; X_i)$ is the mutual information between Y and X_i .

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1: Initialisation: Set  $F \leftarrow \mathbf{X}$  and  $S \leftarrow \emptyset$ 
2:    $f_{\max} = \operatorname{argmax}_{X_i \in \mathbf{X}} I(Y; X_i)$ 
3:   Set  $F \leftarrow F \setminus \{f_{\max}\}$  and  $S \leftarrow f_{\max}$ 
4:   Repeat until  $|S| = K$ :
5:      $f_{\max} = \operatorname{argmax}_{X_i \in F} I(Y; X_i) - \beta \sum_{X_s \in S} I(X_s; X_i)$ 
6:     Set  $F \leftarrow F \setminus \{f_{\max}\}$  and  $S \leftarrow f_{\max} \cup S$ 
7:   End
8: End

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Table 1: Pseudocode of Mutual Information based Feature Selection Algorithm.

[3 marks]

5 marks for question not valid - all questions must have the same number of marks